

Bond performance of GFRP bars with mechanical steel anchors using RILEM beam bond tests: Experimental and theoretical study

Mehran Rahimi^a, Mohammad-Reza Davoodi^a, Mahdi Nematzadeh^{b,*}, Hossein Yousefpour^a

^a Faculty of Civil Engineering, Babol Noshirvani University of Technology, Babol, Iran

^b Department of Civil Engineering, Faculty of Engineering and Technology, University of Mazandaran, Babolsar, Iran

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ABSTRACT

In general, there is a scarcity of practical information currently accessible regarding the bonding characteristics of FRP bars within concrete, and existing guidelines for the design and construction of concrete structures reinforced with FRP, such as ACI 440.1R-15, do not provide equations for different FRP bars, especially when mechanical anchors are used along the bar length. Moreover, the currently available data are from pull-out tests, the reliability of which has been questioned due to unrealistic stress conditions. This paper examines the bond behavior of GFRP bars equipped with mechanical steel anchors through a series of tests on beam specimens. Variables include the diameter of GFRP bars and the compressive strength of concrete as well as the length and thickness of the anchors. The main output parameters under consideration are the maximum transferred stress in the bar, the initial slope of the transferred stress-slip graph, and stress at the initiation of the slip. All investigated steel anchors with different lengths and thicknesses were observed to improve the bond behavior. The rise in the maximum transferred stress due to using anchors ranged between 15% and 60%, depending on the compressive strength of the concrete, GFRP bar diameter, and length and thickness of the steel anchor. The stress at the initiation of the bar slip and the initial bond stiffness were also noticeably greater in specimens containing the anchor compared to those without the anchor. Increasing the anchor length was more effective than the anchor thickness in improving the bond behavior. Based on the experimental results, equations were proposed to predict the values of stress at the first peak of the transferred stress-slip graphs and maximum transferred stress in the bar as a function of the anchor length, anchor diameter, compressive strength of concrete, and diameter of the GFRP bar, which show satisfactory performance in comparison with experimental data.

1. Introduction

The use of fiber-reinforced polymer (FRP) bars as reinforcement in concrete has been on the rise in recent years due to advantages such as their light weight and resistance to corrosion. Various types of FRP bars based on the fiber type include aramid FRP (AFRP), glass FRP (GFRP), basalt FRP (BFRP), and carbon FRP (CFRP) [1–5]. Due to their low cost, GFRP bars are among the most widely used types of FRP bars [5–9].

An important prerequisite for the successful use of FRP reinforcing bars is ensuring a sufficient bond between these bars and concrete, which leads to the transfer of force between the two materials. Bond is particularly important in tensile regions, where concrete cannot carry tensile stresses, and bond failure may lead to the slip of the bar against concrete and the loss of the reinforcing role of the bar [6,10].

According to ACI 440.1R-15 [11], the mechanisms contributing to

the bond between FRP bars and concrete are generally similar to those between steel bars and concrete, which include chemical adhesion between the bar and concrete, friction between the bar surface and surrounding concrete, and the interlocking of the bar deformations with concrete. Whenever the bond level provided by these three factors is not enough, strategies such as bending the bar or the addition of a mechanical anchor to the bar are used [12–17]. The first two mechanisms usually have a limited effect and disappear at relatively small loads; however, the third mechanism provides the most effective contribution [10,12,13].

FRP bars lack strong ribs to develop significant mechanical interlock with concrete. Therefore, in similar conditions, these three mechanisms provide a weaker bond for FRP bars in comparison with steel bars [18–23]. In most cases, bond failure between deformed FRP bars and concrete occurs due to the breaking of the bar rib or the breaking of the

* Corresponding author.

E-mail address: m.nematzadeh@umz.ac.ir (M. Nematzadeh).

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Table 1
Information regarding the beam specimens.

Test specimen designation	Bar diameter (mm)	Concrete compressive strength (MPa)	Number of anchors	Anchor length (mm)	External diameter of anchor (mm)	Internal diameter of anchor (mm)	Anchor thickness (mm)	Number of specimens
C25D8	8	25	0	-	-	-	-	2
C25D8-1A25L20	8	25	1	20	25	20	2.5	2
C25D8-1A30L20	8	25	1	20	30	20	5	2
C25D8-1A25L40	8	25	1	40	25	20	2.5	2
C25D8-1A30L40	8	25	1	40	30	20	5	2
C45D8	8	45	0	-	-	-	-	2
C45D8-1A25L20	8	45	1	20	25	20	2.5	2
C45D8-1A30L20	8	45	1	20	30	20	5	2
C45D8-1A25L40	8	45	1	40	25	20	2.5	2
C45D8-1A30L40	8	45	1	40	30	20	5	2
C25D12	12	25	0	-	-	-	-	2
C25D12-1A25L20	12	25	1	20	25	20	2.5	2
C25D12-1A30L20	12	25	1	20	30	20	5	2
C25D12-1A25L40	12	25	1	40	25	20	2.5	2
C25D12-1A30L40	12	25	1	40	30	20	5	2
C45D12	12	45	0	-	-	-	-	2
C45D12-1A25L20	12	45	1	20	25	20	2.5	2
C45D12-1A30L20	12	45	1	20	30	20	5	2
C45D12-1A25L40	12	45	1	40	25	20	2.5	2
C45D12-1A30L40	12	45	1	40	30	20	5	2

bar rib together with a thin layer of resin (consisting of resin and fibers) from the bar surface [14]. Another factor adding to the complexity of the use of FRP bars is that these bars are usually made of thermosetting resins; thus, if a bent FRP bar is needed, it should be manufactured with the bend in the factory [24]. In other words, after the completion of the production process for FRP bars, they cannot be easily bent, due to which the use of hooked FRP bars may face significant practical limitations.

Several previous studies investigated the bond behavior of FRP bars in concrete [25–28]. The studies revealed that, as the anchorage length increases, the mean maximum shear stress is shown to decrease, due to the nonlinear distribution of the bond stress over the larger area [27, 29–31]. Experimental results from studies show that similar to steel bars, the bond strength of FRP bars declines with an increase in the bar diameter [18,22,29,30]. Hao et al. [32] and Tighiouart et al. [30] stated that when the bar diameter increases, the likelihood of porosity around the bar increases, which lowers the bond strength. Lee et al. [33] showed that greater bond strength is observed in GFRP bars embedded in concrete with higher compressive strength. Other experimental results have demonstrated that the bond strength for FRP bars is correlated with the square root of the concrete compressive strength, similar to that for steel bars [22,25,34].

Given the comparatively modest bond strength of FRP bars, previous studies [11,33,35,36] have aimed at increasing the pull-out resistance of FRP bars through techniques such as the installation of anchors along the FRP bars. Li et al. [37] proposed an anchorage system to improve the bond characteristics of FRP bars and used pull-out tests to assess the effectiveness of the proposed anchor. Mechanical anchors in the form of aluminum tubes with an external diameter of 20 mm and internal diameter of 12 mm were placed on FRP bars with a diameter of 10 mm by applying a radial force. Variables under consideration included the FRP bar type (CFRP and BFRP), bond length (50, 100, and 150 mm), and the number of anchors (zero, one, two, and three). The proposed anchors changed the bond mechanism between the concrete and FRP bar and increased the transferred stress in the bar. In addition, increasing the number of anchors increased the transferred stress in the bar. With a bond length of 150 mm, using two and three anchors increased the transferred stress by 31% and 42% compared with the case without anchors, respectively.

Ashrafi et al. [38] conducted pull-out tests to examine a new anchorage system for GFRP bars consisting of the carbon fiber mat and epoxy resin. Variables included the diameter of the proposed anchorage

(14 and 17 mm) and the concrete compressive strength (16, 24, and 37 MPa). The results demonstrated that the proposed anchorage works better in higher-strength concrete and increased the maximum transferred stress by 7% and 21% in 24 and 37 MPa concrete, respectively, compared with the bar without the anchor. An equation was proposed for the transferred stress in the GFRP bar as a function of bar slip.

In general, there is a scarcity of practical information currently accessible regarding the bonding characteristics of FRP bars within concrete, and existing guidelines for the design and construction of concrete structures reinforced with FRP, such as ACI 440.1R-15 [11], do not provide equations for different FRP bars, especially when mechanical anchors are used along the bar length. Moreover, the currently available data are from pull-out tests, the reliability of which has been questioned due to unrealistic stress conditions. In a pull-out test, the concrete experiences compression while steel experiences tension, which is different from the conditions of a beam, in which both the bar and surrounding concrete are in tension [10].

To overcome the discrepancies in pull-out tests, beam tests have been developed, which represent the bond behavior between the bar and concrete in a more realistic way. In a beam specimen, inclined cracks may form due to the combined action of flexure and shear, similar to the conditions in a real-world structure [10].

The present study investigates the bond behavior of deformed GFRP bars equipped with mechanical steel anchors in concrete through a series of tests on beam specimens according to RILEM RC5 standard [39]. The effects of the compressive strength of concrete (25 and 45 MPa), diameter of the GFRP bar (8 and 12 mm), anchor length (20 and 40 mm), and anchor thickness (2.5 and 5 mm) are evaluated. The results are then used to develop an equation to determine the stress at the first peak of the transferred stress-slip graph and the maximum stress transferred in GFRP bars.

2. Research significance

Examining the literature indicates that FRP bars typically demonstrate lower bond strength in comparison to steel bars, resulting in premature failure of the bond between the bar and concrete. In instances where the bond provided by chemical adhesion, friction, and interlocking is insufficient, strategies like bar hooking and the installation of mechanical anchors are employed. However, the rigidity of FRP bars after production limits the effectiveness of hooking, prompting the proposal of attaching mechanical anchors along the bar length to

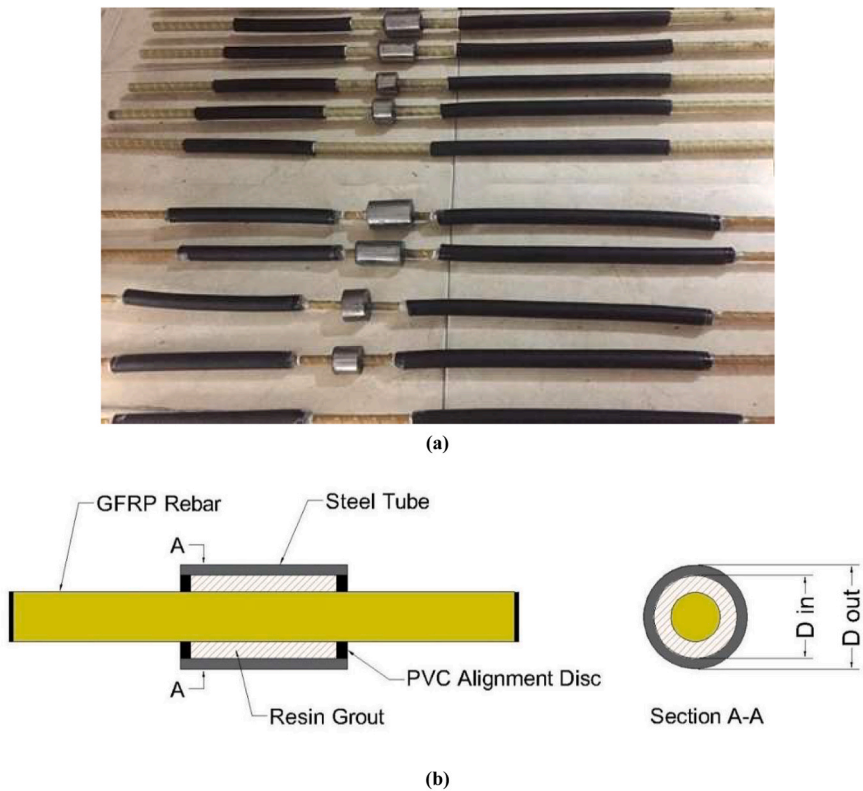


Fig. 1. (a) GFRP bars with anchors; (b) details of the attachment of steel anchors to the GFRP bar.

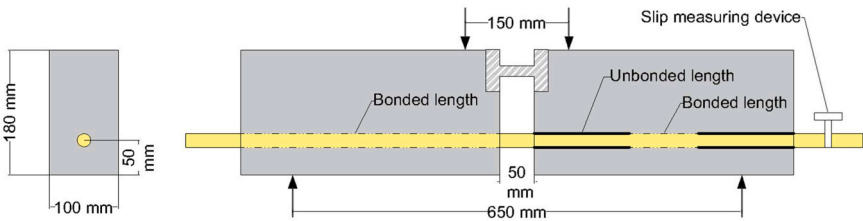


Fig. 2. Beam test according to RILEM RC5 [39].

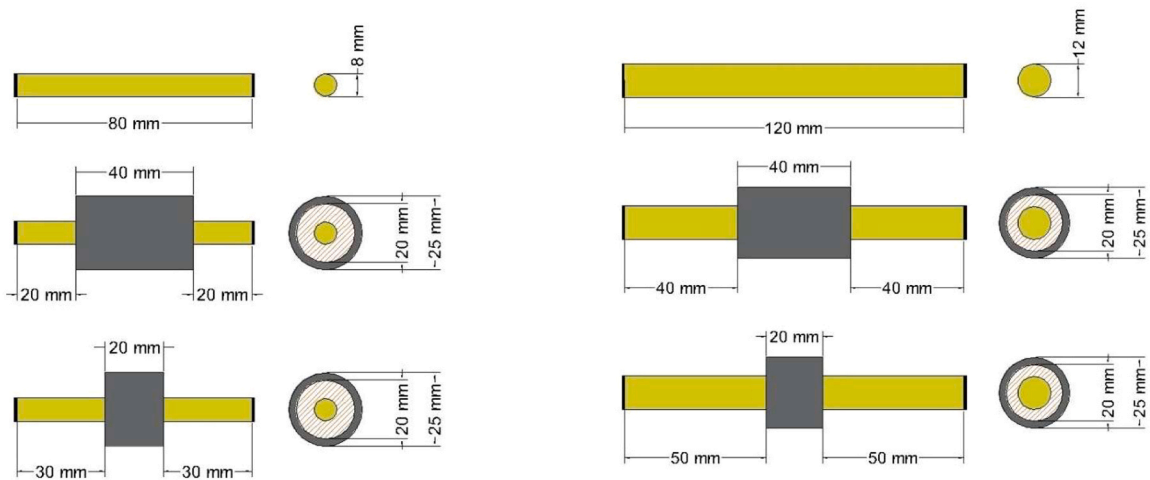


Fig. 3. Dimensions and placement of anchors on the GFRP bars with (a) 8 mm diameter, and (b) 12 mm diameter.

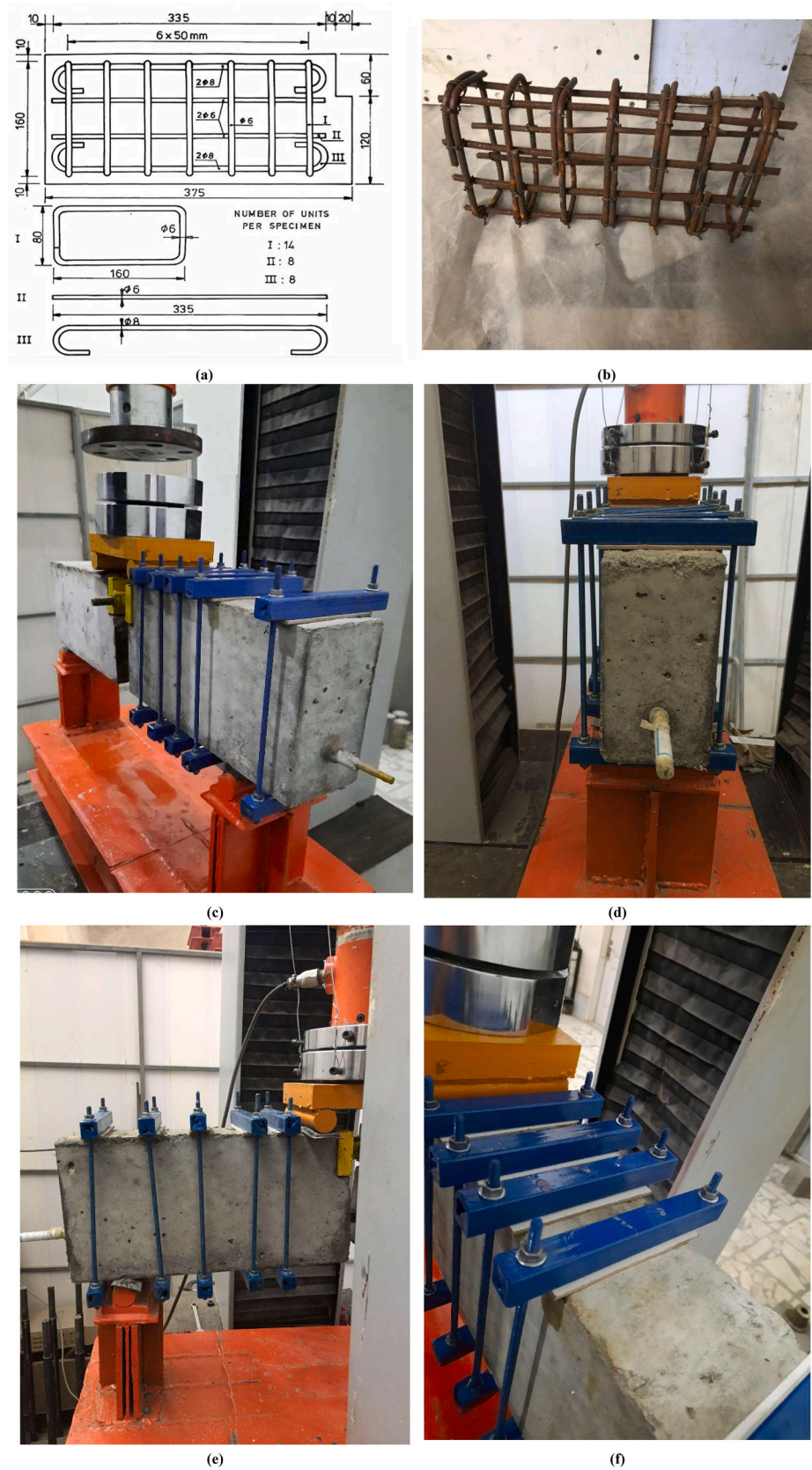


Fig. 4. (a) Details of the reinforcement in the main block according to RILEM RC5 [39]; (b) prepared reinforcement mesh; (c) external stirrups placed in the secondary block; (d), (e), and (f) different views of external stirrups.



Fig. 5. (a) Specimen curing operation; (b) wrapping wet burlap around the specimens and protecting it with a layer of plastic sheet.

Table 2
Characteristics of deformed GFRP bars.

Diameter (mm)	Rib spacing (mm)	Rib height (mm)	Tensile strength (MPa)	Elastic modulus (GPa)
8	10	0.6	1130	47
12	10	0.6	1050	47

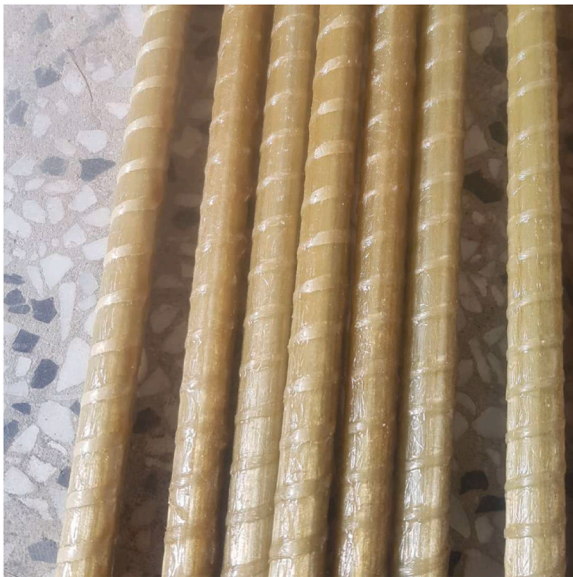


Fig. 6. Deformed GFRP bars used in the specimens.

Table 3
Mechanical properties of steel bars used to fabricate the anchors.

Mechanical properties	Value
Ultimate tensile strength	360 MPa
Tensile yield strength	240 MPa
Elongation at break	20%
Modulus of elasticity	200 GPa

enhance stress transfer and improve bond behavior. Currently, practical information on the bonding characteristics of FRP bars in concrete is scarce, and existing design guidelines, such as ACI 440.1R-15, lack equations for FRP bars with mechanical anchors. On the other hand, many previous studies relied on pull-out tests, which may not accurately simulate conditions experienced by reinforced concrete members. While

pull-out tests are more convenient, the bond strength results may not realistically represent beam behavior. This research aims to evaluate the bond behavior of GFRP bars with varying lengths and thicknesses of steel anchors embedded in concrete through beam specimen tests and develop an equation to determine stress at the first peak and maximum transferred stress in the GFRP bar with the presence of the steel anchor.

3. Research experiment plan

3.1. Specimen description

In the present research, 40 beam specimens according to Type A described in RILEM RC5 [39] were fabricated. The specifics of the specimens in the test program are provided in Table 1. The variables among the specimens included the GFRP bar diameter (8 and 12 mm), concrete compressive strength (25 and 45 MPa), steel anchor length (20 and 40 mm), and steel anchor thickness (2.5 and 5 mm). The internal diameter of all the anchors was 20 mm, and the external diameters of the anchors with thicknesses of 2.5 and 5 mm were 25 and 30 mm, respectively. Per each set of test parameters, two identical specimens were fabricated. The details of the GFRP bars along with the steel anchors utilized in the test specimens are depicted in Fig. 1. The steel anchors, which are visible in the figure, were affixed to the GFRP bars using epoxy resin.

One of the purposes of the present study was to derive an equation to predict the bond performance improvement of GFRP bars in concrete in the presence of steel anchors according to crucial affecting parameters. Hence, anchor length, anchor thickness, GFRP bar diameter, and concrete compressive strength were selected as variables. Since concrete with a compressive strength in the range of 20 to 50 MPa is used in most practical projects around the world, thus, two different levels of 25 and 45 MPa were chosen for concrete compressive strength in the present study. In order to accomplish the beam tests according to type A of the RILEM RC5 [39], a bar diameter smaller than 16 mm should be used; hence, bar diameters of 8 and 12 mm were chosen. It is noteworthy that type A specimens of RILEM RC5 [39] standard have smaller dimensions than type B specimens and are easier to work with. The length and thickness of the steel anchors used in the present study were also selected according to previous research [37,38,40] as well as conducting preliminary tests for the current study.

The specimen names in Table 1 consist of the following numbers: the average cylindrical compressive strength of concrete in MPa, which comes after the letter C; the diameter of the GFRP bar under study, which comes after the letter D; the number of steel anchors, which appears after the hyphen; the outer diameter of the anchor in mm, which is shown after the letter A; and the length of the anchor in mm, which appears after the letter L. Note that the inner diameter of all anchors is the same, equal to 20 mm. The thickness of the anchor is equal to half of the difference between the outer diameter and the inner diameter. For



Fig. 7. Steel anchors used in the specimens.

Table 4

Characteristics of epoxy resin.

Description	Value
Color	Concrete grey (when mixed)
Density (at 25 °C)	1.65 kg/l (when mixed)
Bonding strength	> 3.5 MPa (concrete failed)
Compressive strength	95 MPa (at 35 °C, 7 days)
Tensile & Flexural strength	> 30 MPa
Shear strength	20 MPa
Full cured	After 7 days (at 25 °C)
Working time / pot life	60 min (at 25 °C)

Table 5

Concrete mixture proportions (kg/m³).

Mix ID	Water	Cement	Fine aggregate	Coarse aggregate	High-range water reducer.
C25	195	297	805	1059	1
C45	197	436	663	1059	2.18

Table 6

Measured compressive capacity of the cylindrical specimen.

Mix ID	Compressive strength (MPa)	Average compressive strength (MPa)	COV (%)
C25	22 24.8 23.3	23.4	6
C45	47.4 46.6 44.7	46.2	3

example, Specimen C25D8–1A25L20 is made of 25 MPa concrete, contains a GFRP bar with a diameter of 8 mm, has one anchor with an outer diameter of 25 mm (hence, a thickness of 2.5 mm) and a length of 20 mm.

3.2. Specimen design and fabrication

Fig. 2 gives the schematic view of the tested beam specimens. Each specimen consists of two similar concrete blocks joined by a GFRP reinforcing bar on the tension side and a steel hinge on the compression side. One block is referred to as “the main block”, from which the slip of the GFRP bar relative to concrete is measured. In the main block, the GFRP bar is in contact with concrete over a length that is ten times the diameter of the GFRP bar, according to RILEM RC5 [39]. The GFRP bar is not in contact with concrete over the rest of the main block. To eliminate bonds in the unbonded regions, two polyethylene tubes were used in these regions, which covered the GFRP bar and prevented its

contact with concrete. In the other block (non-main block), however, the GFRP bar and concrete were in contact along the entire length of the block, leading to a slight slip in this block. The slip at the hinged cross-section of the non-main block initiates as the load is applied.

The dimensions and layout of the anchors on the GFRP bars in the main block are shown in Fig. 3. In the main block, a mesh of longitudinal and transverse steel bars is embedded according to RILEM RC5 [39]. In the non-main block, external stirrups were used. Each external stirrup consisted of two box sections and two threaded bars such that by tightening the nuts of the threaded bars, the concrete would be confined at the top and bottom by the box sections to prevent shear failure of concrete. Details of the reinforcement mesh in the main block and external stirrups in the non-main block are shown in Fig. 4.

Once the formwork and reinforcing cage were completed, the concrete pouring operation was performed, after which the surface of the specimens was wrapped with plastic sheets. Twenty-four hours later, the specimens were removed from formwork and were cured for 28 days by wrapping the specimens with wet burlap as the first layer and a plastic sheet as the second layer. Fig. 5 exhibits the specimens during the curing operation.

3.3. Material properties

Table 2 gives the characteristics of the deformed GFRP bars used in this research, a photo of which is also provided in Fig. 6. The steel anchors used in this research were fabricated by machining solid steel bars, and their mechanical properties are provided in Table 3. Data in Tables 2 and 3 are reported by the manufacturers. Fig. 7 shows the steel anchors with different lengths and thicknesses. To attach the anchors to the GFRP bars, epoxy resin with characteristics presented in Table 4 was used; the information in this table is reported by the manufacturers.

The concrete mixture proportions used in the test program are presented in Table 5. These mixture proportions were designed primarily according to ACI 211 [41] with the target cylindrical compressive strength values of 25 and 45 MPa. Limestone aggregate with the maximum nominal aggregate size of 19 mm was used, the gradation for which was in accordance with ASTM C33 [42]. The water absorption of coarse and fine aggregates was 0.5% and 0.7%, and the specific gravity of the coarse and fine aggregates in saturated surface dry (SSD) conditions was 2.68 and 2.64, respectively.

The cement used in the concrete was type II according to ASTM C150 [43]. To reach suitable workability, a polycarboxylates-based high-range water reducer with a specific gravity of 1.05 gr/cm³ was also used within the mixture.

For each concrete grade, three cylindrical specimens with a diameter of 100 mm and height of 200 mm were fabricated and tested based on ASTM C39 [44], and the average strength of these three specimens was reported as the compressive strength of concrete. Table 6 gives the measured and average values of the concrete compressive strength, as

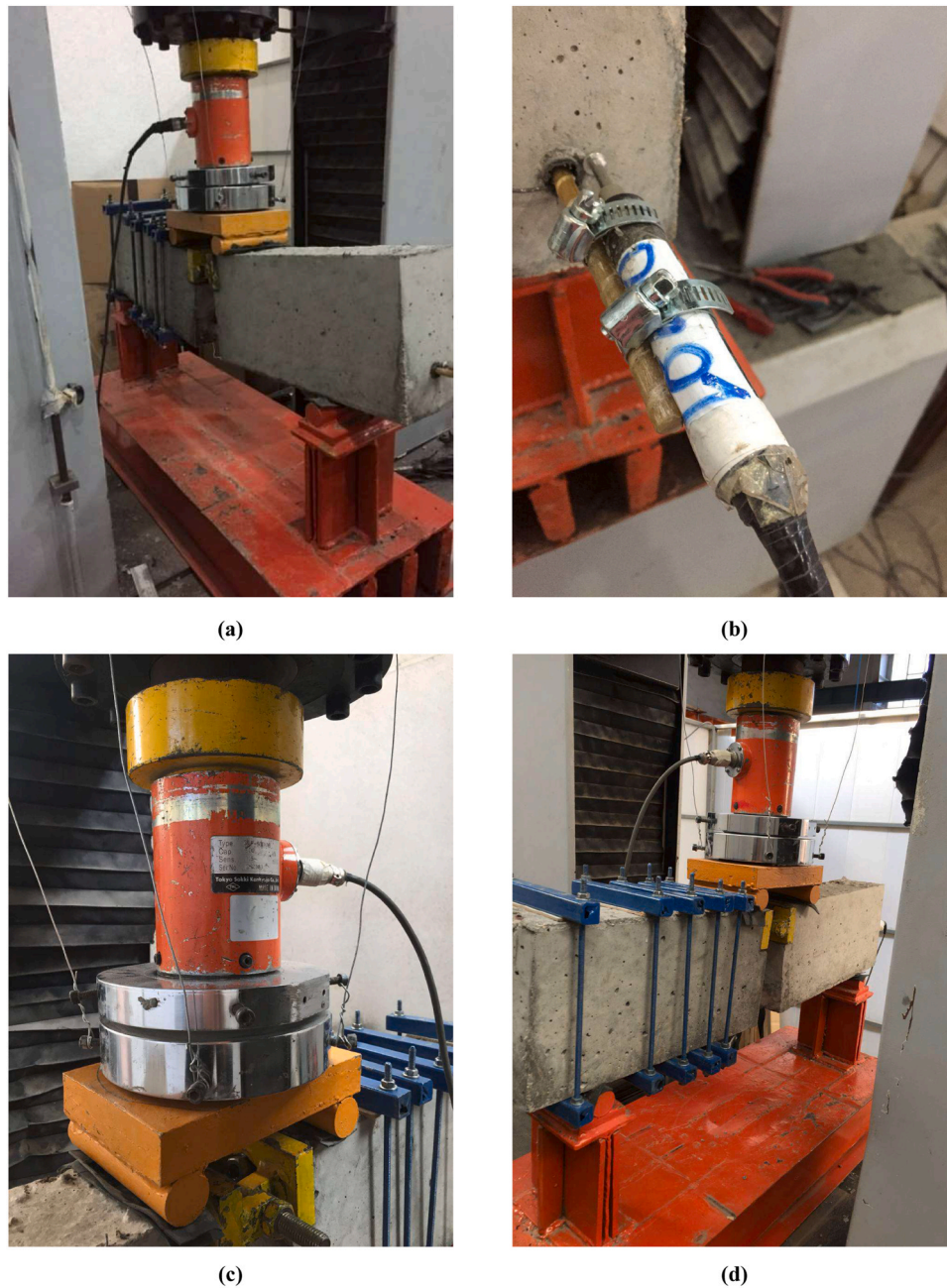


Fig. 8. Photos of the test setup: (a) placement of the beam specimen on the support mounted on the universal machine; (b) position of LVDT on the GFRP bar; (c) the loading plate, spherical heads, and load cell; (d) external stirrups placed in the secondary block.

well as the coefficient of variations (COV) of the results.

4. Test setup and procedure

Photos depicting the details of the setup used to conduct the beam bond test are shown in Fig. 8. A universal testing machine with a load capacity of 400 kN was employed for applying the loads, which were measured via a load cell with a capacity of 300 kN. The slip of the bar against the concrete of the main block was measured using a linear variable differential transducer (LVDT) with a precision of 0.01 mm.

Monotonic loading of the specimens was carried out in a displacement-controlled manner by applying a compressive load to the beam at a constant crosshead displacement rate of 1 mm/min until a minimum slip of 10 mm [45] occurred between the GFRP bar and concrete in the main block. The loading rate was selected based on

RILEM RC5 [39], after converting the rate from load-controlled to displacement-controlled based on preliminary data, which was also consistent with tests conducted by previous researchers [37,38].

It is noteworthy that in general, the slip is not constant along the interface. The slip at the end of the main block is approximately zero up to a certain load, since for small loads the shear stress transfer mainly involves the segment of the interface near the hinge. In addition, the load producing the initiation of the slip at the end of the main block is not an interesting bond property, since it depends on the bonded length within the block [46,47]. However, the test procedure in the present study was carried out according to the recommendations of the RILEM RC5 standard [39] and the test procedure of other research [1,14,30,48–50]. In the RILEM RC5 standard [39], it is clearly mentioned that the slip of the free end of the bar relative to the concrete should be measured. Hence, free-end slip values were used to derive and interpret

Table 7

Peak load and maximum stress results from all specimens.

No.	Specimen name	Beam maximum load (kN)	Maximum tensile stress in GFRP bar (MPa)	Average of maximum tensile stress (MPa)	COV (%)
1	C25D8	22.2	553	528	6.7
2	C25D8-1A25L20	20.2	503		
		25.1	625	619	1.4
		24.6	613		
3	C25D8-1A30L20	28.4	706	714	1.5
		29	721		
4	C25D8-1A25L40	30.7	763	756	1.3
		30	749		
5	C25D8-1A30L40	33.7	839	843	0.6
		34	846		
6	C45D8	24.2	603	605	0.4
		24.3	606		
7	C45D8-1A25L20	28.6	711	723	2.3
		29.5	734		
8	C45D8-1A30L20	29.8	742	738	0.9
		29.5	733		
9	C45D8-1A25L40	32.1	800	832	5.4
		34.7	864		
10	C45D8-1A30L40	32.4	807	847	6.7
		35.6	887		
11	C25D12	44.1	487	509	6
		47.9	530		
12	C25D12-1A25L20	54.3	600	584	4
		51.3	567		
13	C25D12-1A30L20	64.1	709	687	4.5
		60.1	665		
14	C25D12-1A25L40	65.1	720	742	4.1
		69	763		
15	C25D12-1A30L40	69.1	764	787	4.1
		73.2	810		
16	C45D12	47.3	523	559	9.1
		53.7	595		
17	C45D12-1A25L20	63.1	698	696	0.5
		62.6	693		
18	C45D12-1A30L20	64.1	708	704	0.8
		63.3	700		
19	C45D12-1A25L40	76.6	847	824	3.9
		72.4	801		
20	C45D12-1A30L40	77.2	854	842	2
		75.1	830		

the transferred stress-slip graph for the main block, although this may cause insignificant errors. It is better to measure the slip at the bar free-end and the end that is subjected to the load for further studies in the future.

5. Results and discussion

Peak load and maximum stress results obtained from the beam test specimens are provided in Table 7.

For bars without anchors, the bond stress is usually used to compare bond characteristics, which is obtained by dividing the load applied to the bar by the contact surface area of the bar with the surrounding concrete. When anchors are used, however, the transferred stress in the bar, obtained by dividing the force applied to the bar by the cross-sectional area of the bar, is used as the output. The transferred stress in the bar is obtained using Eq. 1 for the Type A specimens according to RILEM RC5 [39]. In this equation, T is the transferred stress in the bar, F

is the load acting on the beam, and A is the cross-sectional area of the GFRP bar.

$$T = \frac{1.25 \times F}{A} \quad (1)$$

In the following sections, the effects of each parameter are investigated separately. Moreover, an equation is proposed to determine the stress at the first peak of the transferred stress-slip graph and the maximum transferred stress in the GFRP bar equipped with the steel anchor based on the parameters considered in this research.

5.1. Failure mode

In this study, likely due to the rebar mesh used in the main block of the beam specimens, all specimens with and without the anchor demonstrated bar pull-out failure mode, and visual inspection showed no trace of cracking at the concrete surface of the specimens.

After completion of the test, some specimens with the anchor were split for further investigation and GFRP bars containing the anchor were extracted from the specimens. There was no sign of failure between the anchors and the GFRP bars, which proves the good performance of the epoxy adhesive applied along the anchors.

5.2. Effect of anchor on the stress-slip behavior

Table 8 shows a comparison between the maximum stress transferred in GFRP bars with and without steel anchors. In this table, the ratios of the maximum transferred stress in specimens containing anchors of

Table 8

Ratio of maximum transferred stress in the specimens containing anchors to that of the reference specimen.

Specimens ID	D8		D12	
	C25	C45	C25	C45
1A25L20	1.17	1.2	1.15	1.25
1A30L20	1.35	1.22	1.35	1.26
1A25L40	1.43	1.38	1.46	1.47
1A30L40	1.6	1.4	1.55	1.51

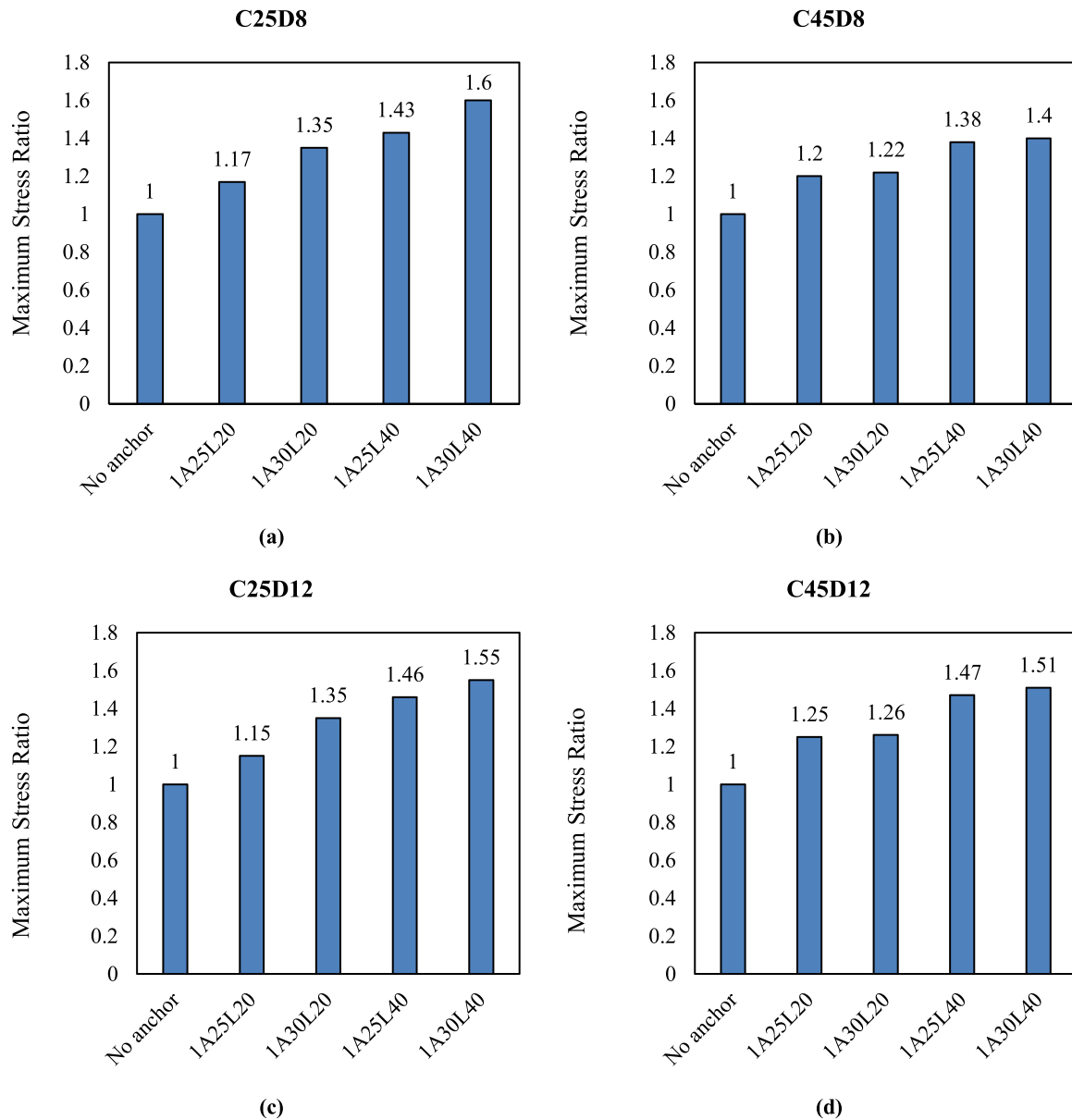


Fig. 9. Ratio of the maximum transferred stress of the specimens to that of the reference specimen. (a) specimen with 25 MPa concrete and 8 mm bar diameter, (b) 45 MPa concrete and 8 mm bar diameter, (c) 25 MPa concrete and 12 mm bar diameter, and (d) 45 MPa concrete and 12 mm bar diameter.

different lengths and thicknesses to that of the specimen without the anchor (regarded as the reference specimen) are provided for the concrete compressive capacity of 25 and 45 MPa and GFRP bar diameters of 8 and 12 mm. A visual presentation of the ratios in Table 8 is also provided in Fig. 9. Graphs showing the variation of the transferred stress in the GFRP bars versus the bar slip relative to the concrete are presented in Fig. 10.

It is seen in Table 8 and Fig. 9 that using steel anchors increased the maximum transferred stress in the bar for all lengths and diameters of anchors, compressive strengths of concrete (25 and 45 MPa), and GFRP bar diameters (8 and 12 mm) investigated in this test program. For the bar diameter of 8 mm, this increase in the maximum transferred stress was 17–60% for the 25-MPa concrete and 20–40% for the 45-MPa concrete. When using the bar with a diameter of 12 mm, this increase is 15–55% and 25–51% for the 25-MPa and 45-MPa concrete, respectively. When using anchors, in addition to the bond stress on the surface area of the bar, a portion of the total load is transferred from the anchor to the concrete, which increases the transferred stress in the bar. The bearing of anchors on the concrete may explain why anchors tend to be

more effective in higher-strength concrete. The maximum increase in the transferred stress occurred in the specimen containing the anchor with a length of 40 mm and thickness of 5 mm, while the minimum increases occurred in the specimen with an anchor measuring 20 mm in length and 2.5 mm in thickness.

Fig. 10 demonstrates that in all specimens with the anchor, the slip initiation occurs at a stress much higher than in the reference specimen. The initial stiffness of all specimens with the anchor is also greater than that of the specimens without the anchor for both bar diameters of 8 and 12 mm and both concrete compressive strengths of 25 and 45 MPa. The maximum transferred stress of all specimens with the anchor is noticeably higher than that of the specimen without the anchor.

The above observations indicate the desirable performance of the steel anchors used in this research. Moreover, it is seen that the improvement of the bond by steel anchors occurs at both concrete compressive strength levels of 25 and 45 MPa.

As indicated by the results in this section, the utilization of steel anchors with varied dimensions in the current study markedly enhances the bond behavior of GFRP bars. For instance, employing a steel anchor

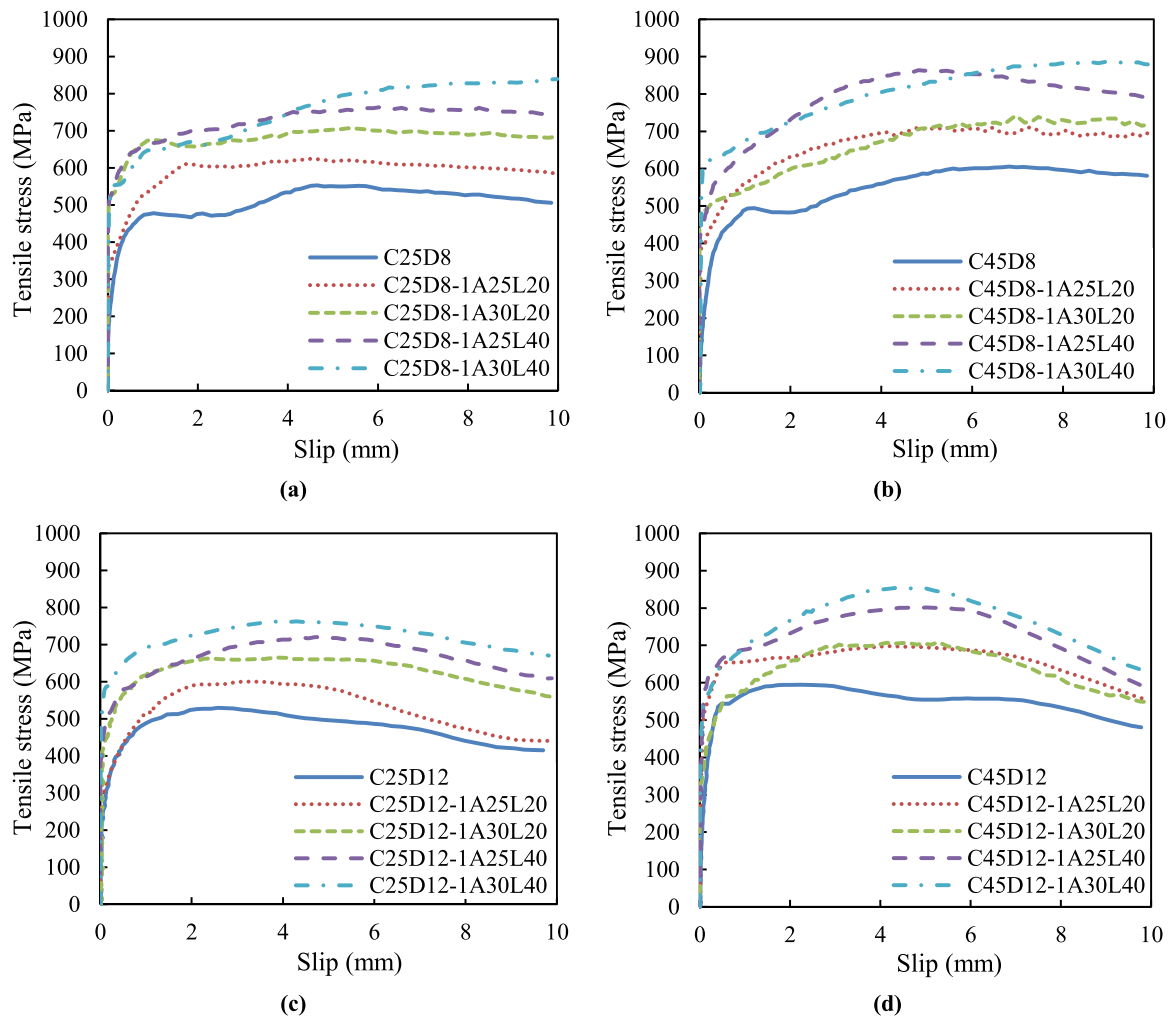


Fig. 10. Transferred stress-slip graph for GFRP bars with different anchors. (a) specimen with 25 MPa concrete and 8 mm bar diameter, (b) 45 MPa concrete and 8 mm bar diameter, (c) 25 MPa concrete and 12 mm bar diameter, and (d) 45 MPa concrete and 12 mm bar diameter.

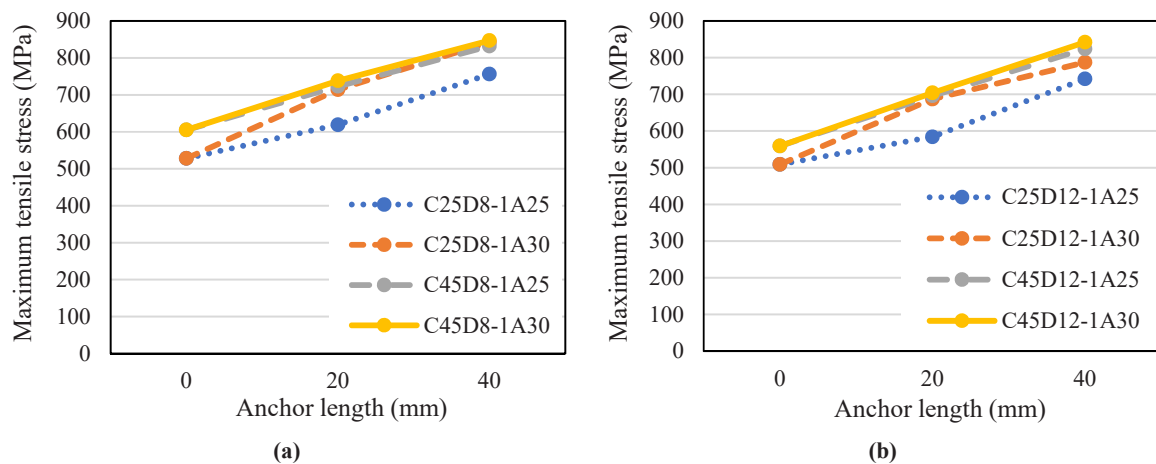


Fig. 11. The maximum transferred stress in the GFRP bars as a function of the anchor length with (a) 8 mm bar diameter, and (b) 12 mm bar diameter.

with a length and thickness of 40 and 5 mm, respectively, resulted in a 60% increase in the maximum stress transferred in an 8 mm GFRP bar diameter compared to the case without an anchor. Similarly, there was a 55% increase observed in a 12 mm bar diameter. In a study conducted by Lee et al. [37], the use of aluminum anchors led to a 0%, 31%, and 42% increase in tensile stress in the GFRP bar for specimens containing

one, two, and three anchors, respectively, with a 150 mm anchorage length. According to Ashrafi et al. [38], utilizing CFRP sheets as anchors resulted in a 21% increase in the maximum tensile stress in the GFRP bar in the best-case scenario compared to the unanchored specimen.

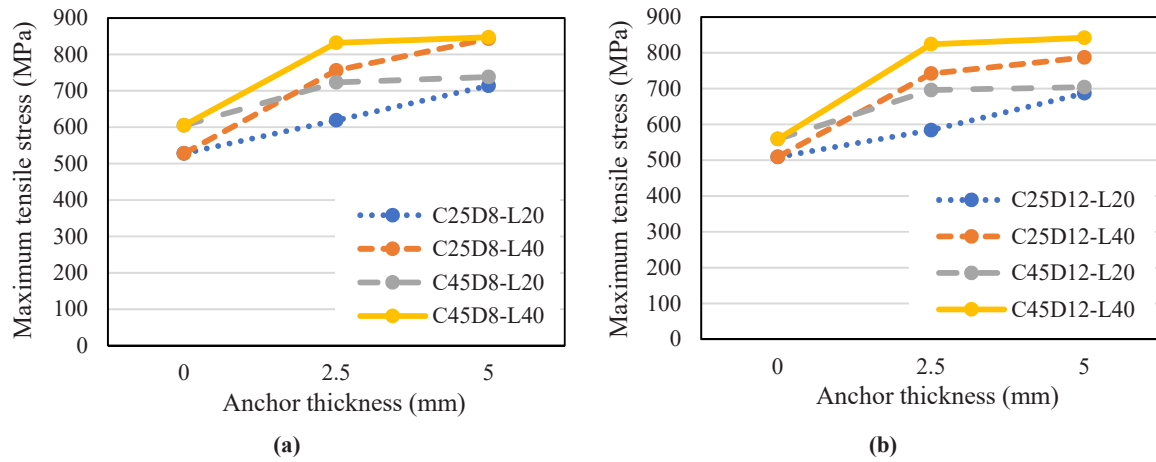


Fig. 12. The maximum transferred stress in GFRP bar as a function of the anchor thickness with (a) 8 mm bar diameter, and (b) 12 mm bar diameter.

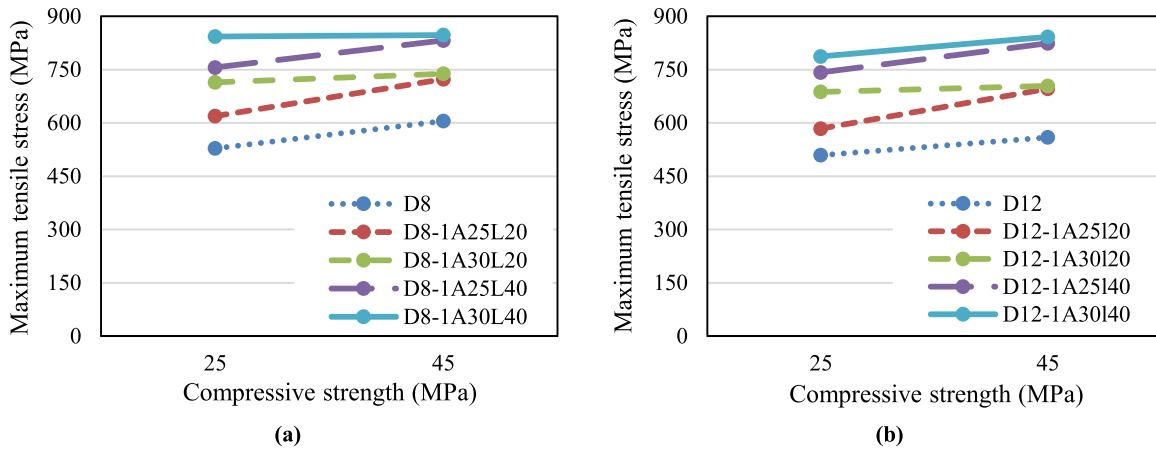


Fig. 13. The maximum transferred stress in the GFRP bar as a function of the compressive strength of concrete with (a) 8 mm bar diameter, and (b) 12 mm bar diameter.

5.3. Effect of anchor length

Linear graphs of the maximum transferred stress in the GFRP bars with different anchors versus the anchor length for the GFRP bar diameters of 8 and 12 mm are illustrated in Fig. 11. As can be seen in Fig. 11, for all GFRP bar diameters, concrete compressive strengths, and anchor thicknesses, increasing the anchor length from 20 to 40 mm led to an increase in the maximum stress transferred in the GFRP bar. For the GFRP bar with a diameter of 8 mm, the increases in the bar stress due to increasing the anchor length were 22% and 18% for anchor thicknesses of 2.5 and 5 mm, respectively, at a concrete compressive strength of 25 MPa and 15% for both thicknesses at a concrete compressive strength of 45 MPa. For the GFRP bar with a diameter of 12 mm, using anchors with thicknesses of 2.5 and 5 mm caused improvements of 27% and 15% in the concrete with the strength of 25 MPa, and 18% and 20% in the concrete with the compressive strength of 45 MPa, respectively.

As can be seen in Fig. 10 for all sets of test parameters, as the anchor length increases from 20 to 40 mm, the slip of the GFRP bar starts to occur at a noticeably higher stress and the stiffness of the graph before the first peak and maximum transferred stress in the bar increase. For example, as seen in Fig. 10a, for the GFRP bar with a diameter of 8 mm containing the anchor with a thickness of 2.5 mm that is embedded in the 25-MPa concrete, the bar slip occurs at a stress of 330 MPa when the anchor length is 20 mm but not until a stress of 530 MPa when the anchor length is 40 mm. In addition, the maximum transferred stress in the bar occurs at a stress of 619 MPa for the anchor length of 20 mm and

756 MPa for the anchor length of 40 mm. When anchors are used, part of the force is carried by the anchor's bearing on the concrete, which is transferred to the bar through the epoxy resin used for attaching the anchor. As the anchor length increases, the contact length of the anchor and the bar and, consequently, the quality of the bond between the anchor and the bar improves, hence leading to increased stresses.

5.4. Effect of anchor thickness

Linear graphs of the maximum transferred stress in the GFRP bar with different anchors are provided in Fig. 12 for the bar diameters of 8 and 12 mm. The figure shows that when using 25-MPa concrete, increasing the anchor thickness from 2.5 to 5 mm increases the transferred stress by 15% and 12% for anchor lengths of 20 and 40 mm on 8-mm GFRP bar and by 18% and 6% for the 12-mm GFRP bar, respectively.

For the 45-MPa concrete, the increase in the maximum transferred stress due to the increase in anchor thickness was limited to 2%. However, for both GFRP bar diameters and anchor lengths of 20 and 40 mm, the maximum transferred stress in the GFRP bar was greater when using 45-MPa concrete compared to the 25-MPa concrete.

Similar to previous observations regarding the effect of anchor length, Fig. 10 indicates that increasing the anchor thickness from 2.5 to 5 mm improved the bond behavior in terms of the stress corresponding to slip initiation as well as the stiffness of the graphs before the first peak and the maximum transferred stress in the bar. Generally, these

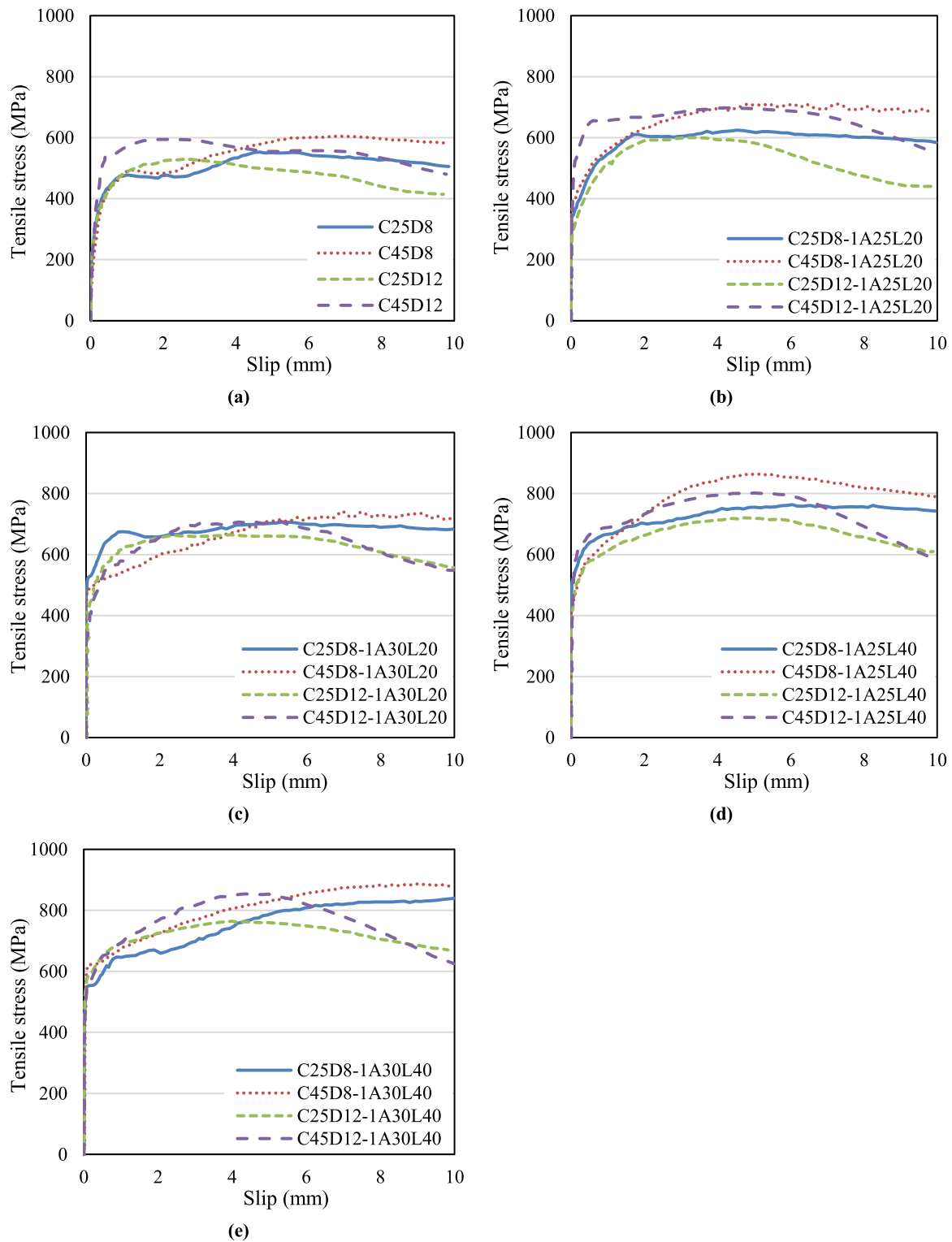


Fig. 14. Exploring the influence of concrete compressive strength on transferred stress-slip characteristics of GFRP bars. (a) specimen without anchor, (b) with 1A25L20, (c) with 1A30L20, (d) with 1A25L40, and (e) with 1A30L40.

parameters increased with the increase in anchor thickness. However, the effects were less pronounced for higher-strength concrete and longer anchor lengths. For example, when the anchor thickness was increased from 2.5 to 5 mm, negligible improvement was observed in the bond stiffness and maximum stress with 20-mm long anchors in 45-MPa concrete, and in bond stiffness and the stress corresponding to slip initiation with 40-mm long anchors in both 25- and 45-MPa concrete.

The improved bond as a result of increased anchor thickness is likely related to the cross-sectional area of the anchor in contact with the concrete, which carries a share of the load, provided that there is sufficient bond with the GFRP bar along the anchor length.

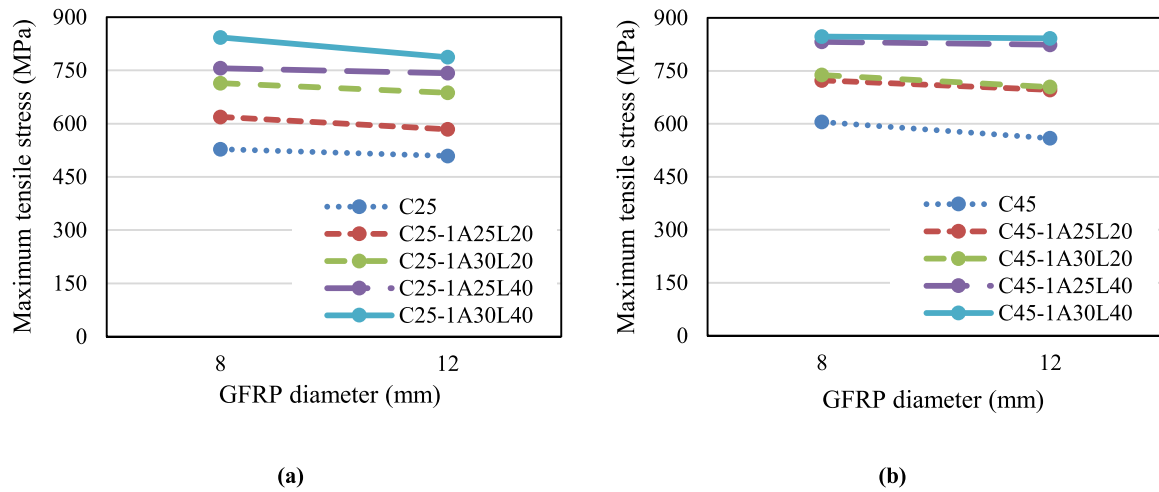


Fig. 15. The maximum transferred stress in the GFRP bar with different anchors as a function of bar diameter with (a) 25 MPa concrete, and (b) 45 MPa concrete.

Table 9

Bond strength of GFRP bars without anchors.

Specimen name	Beam maximum load (kN)	Force in GFRP bar (kN)	L_{bond} (mm)	Bond strength (MPa)	Average of results (MPa)	COV (%)
C25D8	22.2	27.8	80	13.8	13.2	6.4
	20.2	25.3		12.6		
C45D8	24.2	30.3	80	15.1	15.1	0
	24.3	30.4		15.1		
C25D12	44.1	55.1	120	12.2	12.7	5.6
	47.9	59.9		13.2		
C45D12	47.3	59.1	120	13.1	14	8.6
	53.7	67.1		14.8		

5.5. Comparison of effects of increasing the length and thickness of the anchor

The two previous sections demonstrate that to increase the maximum transferred stress in the GFRP bar, it is more effective to increase the anchor length rather than the anchor thickness. In particular, in the concrete with a strength of 45 MPa, increasing the anchor thickness caused only a 2% increase in the maximum transferred stress in the GFRP bar. Moreover, as observed in Fig. 10, in several cases, increasing the anchor thickness from 2.5 to 5 mm results in negligible improvements in the slip-initiation stress, bond stiffness, or maximum transferred stress, all of which were observed to always increase when the anchor length is changed from 20 to 40 mm. The greater effect of anchor length in comparison with anchor thickness indicates the importance of

bond length between the anchor and GFRP bar.

5.6. Effect of concrete compressive strength

Linear graphs of the maximum transferred stress in the GFRP bars with diameters of 8 and 12 mm as a function of the concrete compressive strength are presented in Fig. 13. It is seen that in almost all cases, increasing the concrete strength from 25 to 45 MPa improved the maximum transferred stress in the GFRP bar. In specimens without the anchor, as the strength of concrete was increased from 25 to 45 MPa, the maximum transferred stress increased by 15% and 10% for the 8- and 12-mm bars, respectively. For specimens containing anchors, the increase in the maximum transferred stress due to increasing the concrete compressive strength was between 0% (for the anchor with length and thickness of 40 and 5 mm, respectively, installed on 8-mm bar) and 19% (for the anchor with length and thickness of 20 and 2.5 mm, respectively, installed on 12-mm bar).

Fig. 14 presents the transferred stress-versus-slip curves for 8- and 12-mm bars embedded in 25- and 45-MPa concrete, which are separated based on different anchor conditions. It is seen that in the case without the anchor, increasing the concrete compressive strength from 25 to 45 MPa has no noticeable effect on the stress at the bar slip threshold and stiffness of the graphs before the first peak but increases the maximum transferred stress in the bar.

For all specimens containing the anchor, increasing the compressive strength caused an increase in the maximum transferred stress. However, the stress corresponding to slip initiation increased only for specimens containing the anchor with a length of 20 mm and a thickness of 2.5 mm. For longer or thicker anchors, no noticeable effect on the slip-initiation stress or bond stiffness could be detected because of increasing the compressive strength of concrete.

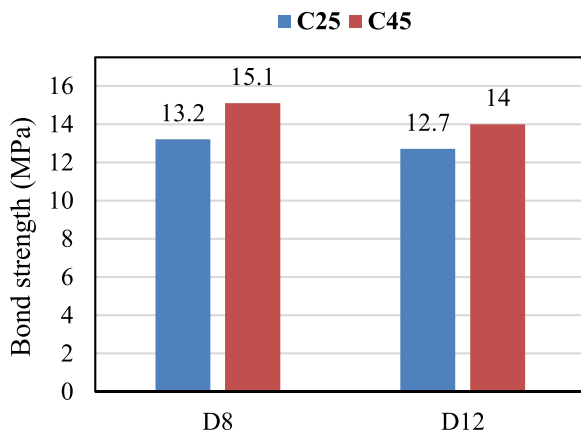


Fig. 16. Bond strength of the GFRP bars without anchors.

Table 10

Comparison of test data with predictions by Eq. 3 and Eq. 4.

No.	Specimen name	T _{first-peak} , experiment (MPa)	T _{max} , experiment (MPa)	T _{first-peak} , predicted (MPa)	T _{max} , predicted (MPa)	Experimental/ Predicted T _{first-peak}	Experimental/ Predicted T _{max}
1	C25D8	475	528	504	532	0.94	0.99
2	C25D8-1A25L20	610	619	617	656	0.99	0.94
3	C25D8-1A30L20	673	714	616	691	1.09	1.03
4	C25D8-1A25L40	672	756	669	769	1.01	0.98
5	C25D8-1A30L40	641	843	668	809	0.96	1.04
6	C45D8	494	605	506	577	0.98	1.05
7	C45D8-1A25L20	654	723	619	712	1.06	1.02
8	C45D8-1A30L20	562	738	618	749	0.91	0.98
9	C45D8-1A25L40	656	832	671	834	0.98	1.00
10	C45D8-1A30L40	715	847	670	878	1.07	0.96
11	C25D12	509	509	507	513	1.00	0.99
12	C25D12-1A25L20	583	584	620	633	0.94	0.92
13	C25D12-1A30L20	655	687	620	666	1.06	1.03
14	C25D12-1A25L40	647	742	673	741	0.96	1.00
15	C25D12-1A30L40	695	787	672	780	1.03	1.01
16	C45D12	544	559	509	556	1.07	1.00
17	C45D12-1A25L20	655	696	623	686	1.05	1.01
18	C45D12-1A30L20	565	704	622	723	0.91	0.97
19	C45D12-1A25L40	693	824	675	804	1.03	1.02
20	C45D12-1A30L40	654	842	674	847	0.97	0.99

5.7. Effect of GFRP bar diameter

Fig. 15 exhibits linear graphs of the maximum transferred stress in the bar as a function of bar diameter. It is observed that in all cases, as the diameter of the GFRP bar increases from 8 to 12 mm, the maximum transferred stress in the GFRP bar slightly declines. For specimens without anchors, this decline is 4% in the 25-MPa concrete and 8% in the specimens with the 45-MPa concrete. A generally similar slight reduction was observed in specimens containing the anchor, which varied between 1% (for the anchor length of 40 mm, anchor thickness of 2.5 mm, and concrete compressive strength of 45 MPa) and 7% (for the anchor length of 40 mm, anchor thickness of 5 mm and concrete compressive strength of 25 MPa).

The stress-slip graphs in Fig. 14 show that in almost all specimens, with or without anchors, increasing the bar diameter from 8 to 12 mm leads to no noticeable change in the stress at the initiation of slip and the stiffness of the graphs before the first peak, but slightly lowers the maximum transferred stress in the GFRP bar.

5.8. Bond strength for specimens with no anchors

In bars without an anchor, the average bond stress is usually used to compare the bond quality in different conditions. The average bond stress is obtained by dividing the force by the contact area of the bar with concrete, as shown in Eq. 2. In this equation, p is the load applied to the bar, d_b is the bar diameter, and L_{bond} is the contact length of the bar with concrete.

In Type A specimens per RILEM RC5 [39], p is $1.25F$, where F is the peak force acting on the beam. The maximum value of the bond stress is referred to as the bond strength, which is presented in Table 9 for the GFRP bars without the anchor. In addition, Fig. 16 presents the bond strength graphs of the 8- and 12-mm bars in concrete with compressive strengths of 25 and 45 MPa.

$$\tau_{ave} = \frac{p}{\pi \times d_b \times L_{bond}} \quad (2)$$

As visible in Fig. 16, the bond strength of GFRP bars was observed to decline as the bar diameter increased. The decline in the bond strength due to an increase in the bar diameter from 8 to 12 mm was approximately 4% for the 25-MPa concrete and 10% for the 45-MPa concrete. This observation is consistent with the observations by previous researchers on steel and FRP bars [18,22,29,30] and is attributed to the increased likelihood of pore formation around larger-diameter bars,

which lowers the contact area between the concrete and the bar and leads to decreased bond strength. However, increasing the compressive strength of concrete from 25 to 45 MPa improves the bond strength of the 8- and 12-mm GFRP bars by 14% and 10%, respectively.

5.9. Proposed equations

Based on the regression analysis of the experimental data obtained in this research using the discrete least squares method, Eqs. 3 and 4 are proposed to predict the stress at the first peak of the transferred stress-slip graph and the maximum transferred stress in the GFRP bar in the presence of the steel anchor, respectively. The coefficients of determination were 0.77 and 0.96, for Eqs. 3 and 4, respectively. In these equations, $T_{first-peak}$ is the stress at the first peak of the transferred stress-slip graph in MPa, T_{max} is the maximum transferred stress in the bar in MPa, f'_c is the compressive strength of concrete in MPa, and d_b is the bar diameter in mm. Furthermore, t_a and l_a are the thickness and length of the steel anchor in mm, respectively. It should be emphasized that the proposed equations remain valid within the specified ranges for the variables used in this research.

$$T_{first-peak} = (478.8f'_c d_b^{0.0144})(1 + l_a^{0.18}) / (1 + 1.214t_a^{0.0036}) \quad R^2 = 0.77 \quad (3)$$

$$T_{max} = 1 + 365f'_c \exp\left(\frac{-0.0091d_b + 0.0207t_a}{0.992l_a}\right) \quad R^2 = 0.96 \quad (4)$$

Table 10 presents a comparison between the experimentally obtained stresses at the first peak and the maximum transferred stresses and those predicted by Eqs. 3 and 4. The last two columns of the table show the ratios of the experimental results to values predicted by the equation, which varied between 0.91 and 1.09 for the stress at the first peak (Eq. 3) and between 0.92 and 1.05 for the maximum transferred stress (Eq. 4). Therefore, the performance of both equations is deemed satisfactory.

Fig. 17 compares the experimental results with predictions of Eq. 3 and presents the impact of various parameters on the predicted values of the stress at the first peak of the transferred stress-slip graph. It is seen in Fig. 17a that the proposed Eq. 3 can properly predict the experimental data on the first peak stresses. Fig. 17b indicates that the predicted values by Eq. 3 are more accurate for 45-MPa concrete than for 25-MPa concrete. In addition, Fig. 17c demonstrates that the accuracy of the predicted results is almost similar for the two bar diameters of 8 and 12 mm. In Fig. 17d and e, it is observed that the predictions are more

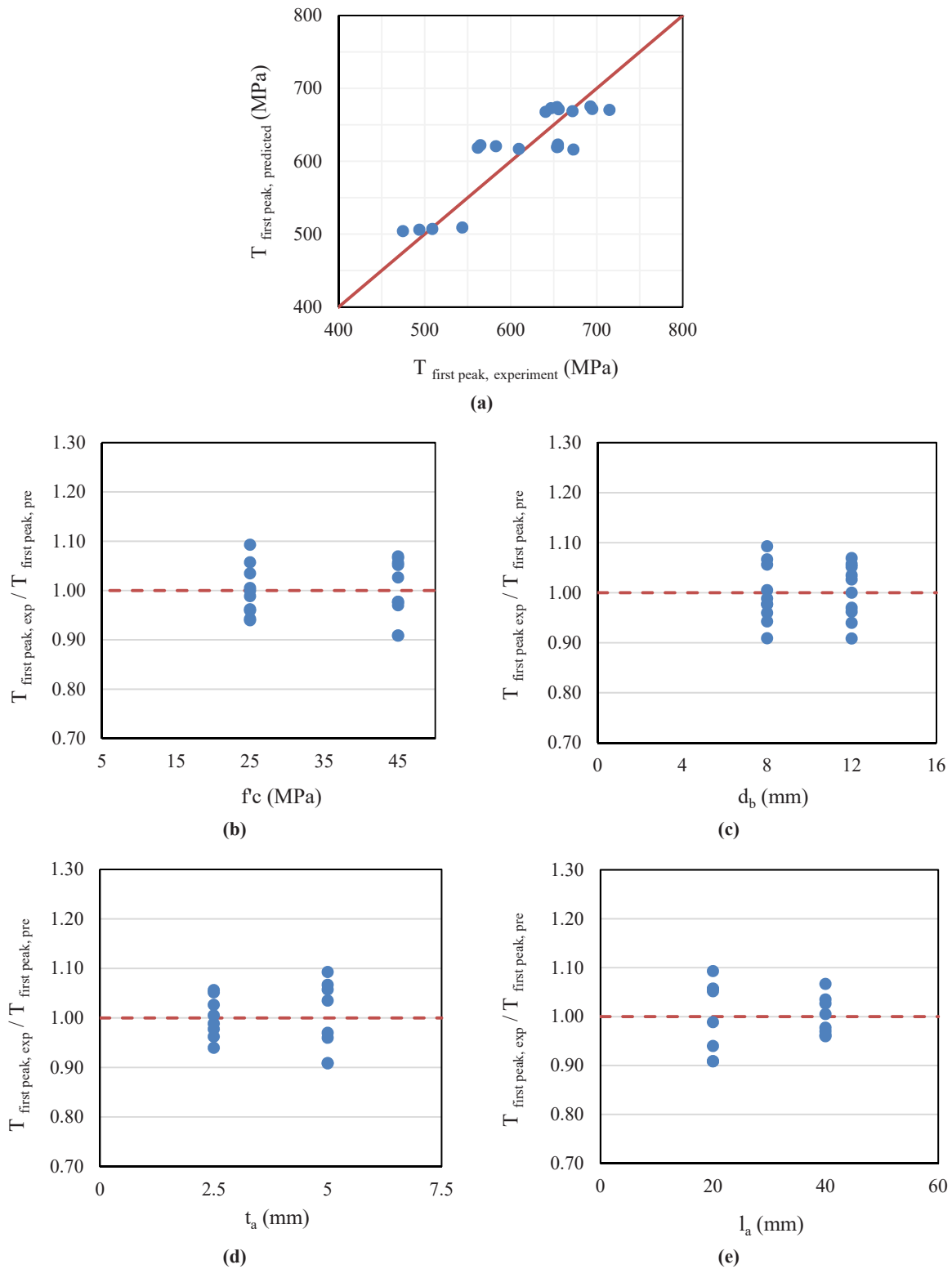


Fig. 17. Experimental vs predicted stress at first peak values and the impact of various parameters on the ratio of experimental stress at first peak to that predicted by Eq. 3; (a) Experimental vs predicted values, (b) impact of concrete compressive strength, (c) impact of bar diameter, (d) impact of anchor thickness, and (e) impact of anchor length.

accurate for specimens with an anchor thickness of 2.5 mm compared to those with an anchor thickness of 5 mm, and for specimens with an anchor length of 40 mm compared to those with an anchor length of 20 mm.

Fig. 18 presents similar graphs comparing the predicted results from

Eq. 4 and the experimental data regarding the maximum transferred stress in the bar. Fig. 18a exhibits a good correlation between the proposed Eq. 4 and the experimental results. Fig. 18b, c and e show that the proposed equation is slightly more accurate for the 45-MPa concrete specimens than for the 25-MPa specimens, for specimens with the 12-

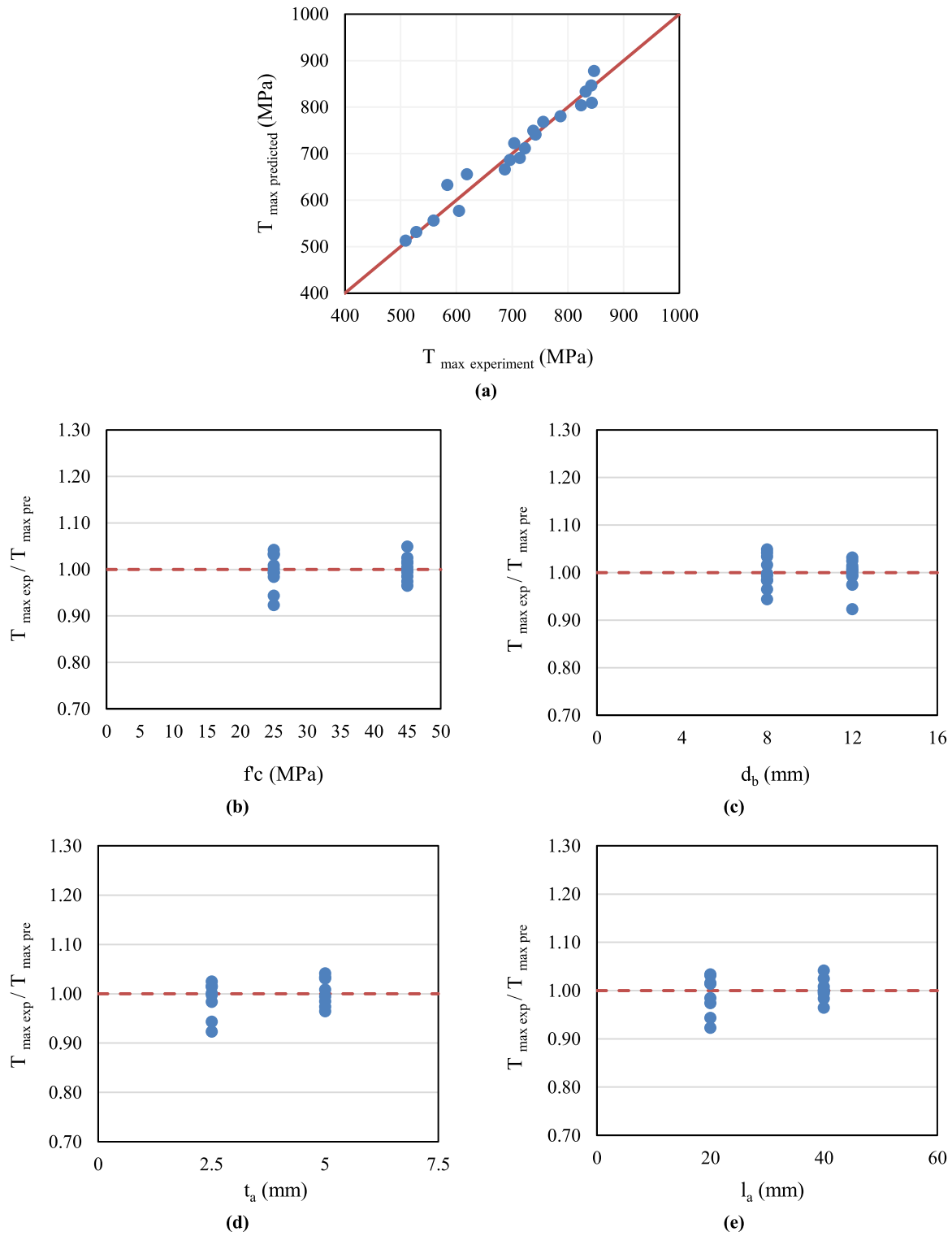


Fig. 18. Experimental vs predicted maximum tensile stress values and the impact of various parameters on the ratio of experimental maximum tensile stress values to that predicted by Eq. 4. (a) Experimental vs predicted values, (b) impact of concrete compressive strength, (c) impact of bar diameter, (d) impact of anchor thickness, and (e) impact of anchor length.

mm diameter bar than those containing the 8-mm bar, and for specimens with an anchor length of 40 mm compared to those with an anchor length of 20 mm. However, as visible in Fig. 18d, the accuracy of predicted values is almost similar for both anchor thicknesses of 2.5 and 5 mm.

Note that at the time of writing this paper, no similar study has

addressed the effect of the steel anchors installed with the method used in this study on the bond performance of the GFRP bar. Therefore, it was not possible to compare the proposed Eqs. 3 and 4 with the experimental data by other researchers.

6. Conclusions

This research was carried out to assess the bond behavior of GFRP bars equipped with steel anchors with different lengths and thicknesses embedded in concrete and develop an equation to determine the stress at the first peak and maximum transferred stress in the GFRP bar in the presence of the steel anchor. This objective was reached by conducting a series of beam bond tests. The main results of this research are listed below.

1. The use of steel anchors with different lengths and thicknesses on the GFRP bars with diameters of 8 and 12 mm in concrete with compressive strength levels of 25 and 45 MPa increased the maximum transferred stress in the bar. In the specimens containing the anchor, the GFRP bar started to slip at a much greater stress, and the bond stiffness before the first peak was noticeably greater compared to the specimen without the anchor.

2. For both bar diameters of 8 and 12 mm, both concrete levels of 25 and 45 MPa, and both anchor thicknesses of 2.5 and 5 mm, increasing the anchor length from 20 to 40 mm led to an increase in the stress corresponding to the first slip, the bond stiffness before the first peak, and the maximum transferred stress in the bar.

3. In general, increasing the anchor thickness from 2.5 to 5 mm improved the bond behavior but to a lesser extent compared to the increase in the anchor length from 20 mm to 40 mm. In most specimens, increasing the concrete compressive strength from 25 to 45 MPa also enhanced the bond behavior.

4. The bond strength of the GFRP bars with or without anchors declines with the increase in the bar diameter. This decline was more noticeable when using 45-MPa concrete compared to 25-MPa concrete.

5. Based on the experimental results, equations were proposed to predict the values of stress at the first peak of the transferred stress-slip graphs and maximum transferred stress in the bar as a function of the anchor length, anchor diameter, compressive strength of concrete, and diameter of the GFRP bar. The proposed equations show a good correlation with the experimental results.

Application of research results

Due to the limitations of FRP bar hooking, the findings from this current study can provide valuable insights into the utilization of mechanical steel anchors with diverse lengths and thicknesses along the length of FRP bars. Consequently, the use of mechanical anchors along the length of FRP bars becomes a preferable option, particularly in situations requiring an extended development length. For example, in side and corner beam-column joints, mechanical anchors can replace hooked bars along the length of the FRP bars. Utilizing the equations presented in this study facilitates the prediction of the impact of anchors with specific lengths and thicknesses on enhancing the bond behavior of GFRP bars.

CRedit authorship contribution statement

Rahimi Mehran: Conceptualization, Formal analysis, Investigation, Methodology, Resources, Writing – original draft, Writing – review & editing. **Davoodi Mohammad-Reza:** Conceptualization, Formal analysis, Methodology, Project administration, Supervision. **Nematzadeh Mahdi:** Conceptualization, Formal analysis, Methodology, Project administration, Supervision, Writing – review & editing. **Yousefpoor Hossein:** Conceptualization, Formal analysis, Methodology, Project administration, Supervision, Writing – review & editing.

Declaration of Competing Interest

We wish to confirm that there are no known conflicts of interest associated with this publication and there has been no significant financial support for this work that could have influenced its outcome.

Data availability

Data will be made available on request.

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